P.1 Functions and Their Graphs

Name _____ Period ____ Date _____

AP Calculus P.1 · Student sheet

I can…

- evaluate a function and its difference quotient.
- find the domain of a function and tell whether it is even, odd, or neither.
- find the average rate of change of a function as the slope of a secant line.

Vocabulary · fill in the missing word

Function — a relation in which each element of the domain _____ to exactly one element of the range

Difference quotient — _____

Even and odd functions — even: $f(-x) = f(x)$ for every x in the _____; odd: $f(-x) = -f(x)$

Average rate of change — the change in y _____ by the change in x, $\frac{f(b) - f(a)}{b - a}$

Part 1 · Secant slopes

$f(x) = x^2$. Slope of the secant line from (1, 1) to (b, f(b)).

b	3	2	1.5	1.1	1.01
f(b)					
$(f(b) - 1)/(b - 1)$					

Part 2 · Notes and examples (fill in each note, then work the example)

Function notation: in $y = f(x)$, x is the _____ variable and y is the dependent variable.

Example 1

$f(x) = 3x^2 + 2x - 4$. Find $f(0)$, $f(-x)$, and $f(x + 1)$.

Domain: unless it is stated, the domain is the set of all real numbers x for which f(x) is a _____ number.

Example 2

Find the domain of each function.

a. $f(x) = \frac{x}{x^2 + 1}$

b. $v(t) = \sqrt{t^2 - 9}$

c. $h(x) = \frac{x + 2}{x^3 - 4x}$

Difference quotient: _____, $h \neq 0$

Example 3

Find the difference quotient of $f(x) = x^2 + 2x$.

Symmetry: an even function is symmetric with respect to the y-axis; an odd function is symmetric with respect to the _____.

Example 4

Even, odd, or neither?

a. $h(x) = \dfrac{x}{x^2 - 1}$

b. $G(x) = \sqrt{x}$

c. $f(x) = \sqrt[3]{(3x^2 + 1)}$

Slope of a secant line: the average rate of change of f from a to b equals the slope of the secant line through (a, f(a)) and

(_____).

Example 5

$f(x) = -2x^2 + 4$. Find the average rate of change of f

a. from 1 to 3

b. from 1 to x, $x \neq 1$

Part 3 · Partner practice

Partner 1

Find the difference quotient of $f(x) = -3x + 1$.

Partner 2

$s(t) = 20 - 0.8t^2$. Find the average rate of change from $t = 1$ to $t = 3$.